

Module MC-3: Spectral features for motion data

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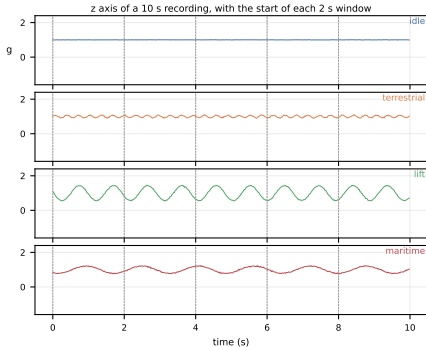
After this module you can:

- ▶ Compute the time-domain features of a window of a motion signal: the root mean square, the skewness, and the kurtosis.
- ▶ Explain what the FFT length changes: the number of bins, and the distance between two bins.
- ▶ Put the time-domain features and the spectral power into one feature vector for the three axes of an inertial sensor.
- ▶ Choose a feature set for a device from a measured comparison, and not from a rule.

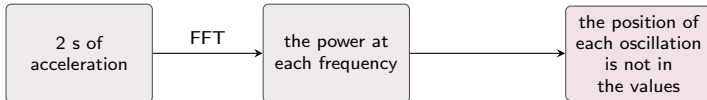
- ▶ An inertial sensor gives 50 samples per second on each of three axes. A window of 2 s gives 300 values.
- ▶ The network must learn by itself that a slow oscillation and a faster one are two different situations. It needs many windows to learn that.
- ▶ Signal processing gives the network the same information in fewer values, and the meaning of each value is known before the training starts.

Key point: Every value that the network reads has a name and a unit.

Cut the recording into windows



- ▶ A window of 2 s, a new window every 0.2 s. Each window is one row of the dataset.
- ▶ The window must hold a few cycles of the motion. A window shorter than one cycle hides the frequency.
- ▶ The task of the lab is the transport of a container. It rests in a warehouse (*idle*), rides a truck or a train (*terrestrial*), is moved by a forklift (*lift*), or sails (*maritime*).



- ▶ The network sees the shape of the signal. It has no way to ask which frequency the shape contains.
- ▶ The RMS of a window says how big the motion is. It says nothing about how fast it is.
- ▶ Two motions with the same size and different speed give the same RMS and different spectra.

- ▶ The kit lies with the display up, so the z axis shows gravity. Its value is near $+1$ g whatever the motion is.
- ▶ A large constant shifts every feature and hides the small changes that carry the class.
- ▶ Subtract the mean of the window from each sample. Bin 0 of the FFT is then zero, and the answer can start at bin 1.

Key point: The RMS before the removal of the mean and after it are two different numbers.

$$x_{\text{RMS}} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}$$

- ▶ One value for the size of the motion. It has the unit of the signal, so it stays in g.
- ▶ It is the value of a constant direct current that would give the same power in a resistor.
- ▶ It is the cheapest feature of the block, and it is in every version of it.

The formulas in this module use no correction for the sample size. Then a laptop and a board give the same number.

- ▶ Skewness measures the asymmetry of the distribution of the values of a window. It is below zero when the long tail is on the left.
- ▶ Kurtosis measures how heavy the tails are. A normal distribution has 0.
- ▶ A value far from 0 says that the motion is not a smooth oscillation. It is a shock or a knock.

Signal	Skewness, kurtosis
Smooth sine wave	near 0, near -2
One shock in the window	large, large
Sensor noise	near 0, near 0

The kurtosis of a sine wave is -1.5 , and the sample of a normal distribution is near -2 , because the module subtracts 3.

- ▶ Two definitions of the skewness exist: one with a correction for the sample size, and one with no correction.
- ▶ For the z axis of the window of the project of the chapter, the Studio reports -0.0978 , 0.1735 , and 6.8629 . The same values with the correction are -0.0990 , 0.1756 , and 6.9463 .
- ▶ The two differ by about one percent on the third axis. A model trained on the values of one definition and run on the values of the other one loses accuracy.

Key point: Write the formula in the report. A number without a formula is not a feature.

Source: Values reported in "Machine Learning Systems", chapter "DSP Spectral Features", for the window of the project.

Feature	What it measures	Version
RMS	the size of the motion	1 and 2
Skewness	the asymmetry of the values	2
Kurtosis	the weight of the tails	2

- ▶ Three values per axis. The first version of the block computed only the RMS, plus the peaks and the power of the FFT, for 11 values per axis and 33 values in total.
- ▶ The second version adds the skewness and the kurtosis, and the studio then reports 39 values, so 13 per axis.

Source: Feature lists of "Machine Learning Systems", chapter "DSP Spectral Features", versions 1 and 2 of the Spectral Analysis block.

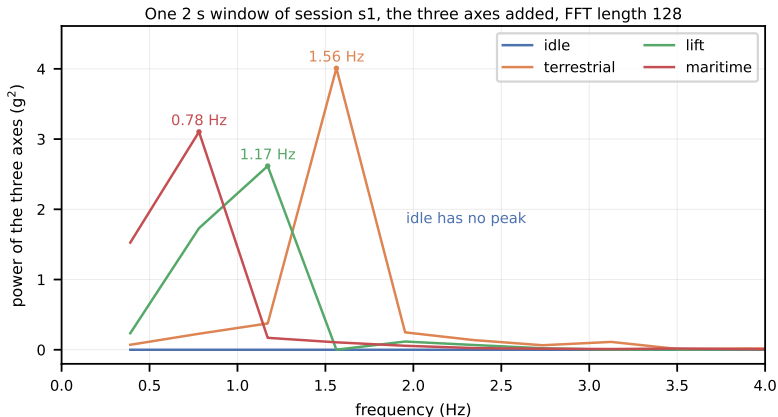
The FFT turns samples into bins

$$\text{distance between two bins} = \frac{f_s}{N} \quad \text{resolution} \approx \frac{1}{T}$$

- ▶ f_s is the sampling rate, N is the FFT length, and T is the length of the window.
- ▶ A window of 2 s at 50 Hz has 100 samples. An FFT of length 128 pads it with 28 zeros. The bins then sit every $50/128 = 0.39$ Hz, and no bin is closer than the 0.5 Hz that 2 s of signal can resolve.
- ▶ Padding adds no information. It only draws the same peak on a finer grid.

Key point: More bins means a finer grid, not a better sensor.

The power at each frequency



Each class that moves has a band of its own, and `idle` has no band because it does not move. The rate of a class follows the speed of its session, so two bands can touch. The network sees 64 numbers per axis and no shape at all.

Source: Figure drawn from the simulated signals of the lab notebook, with the power of the three axes added. The notebook plots one axis at a time. The signals are simulated, so the figure is an illustration.

- ▶ A window has more samples than the FFT length. So the window is cut into frames, and the FFT runs on each frame.
- ▶ Welch's method averages the power of the frames. The estimate is smooth, and a peak that lasts a short time is reduced.
- ▶ The block of the studio keeps the largest power of each bin over all the frames. The function in the chapter divides the signal into frames, takes the power of each frame, and keeps the maximum.
- ▶ Max-hold keeps a short event. Welch removes it. Choose with the motion in mind: a pallet that hits once in a window is an event.

Source: Description and function of "Machine Learning Systems", chapter "DSP Spectral Features", section "Spectral features".

The feature vector of one axis

Group	Values	FFT length
RMS, skewness, kurtosis	3	any
Spectral power of bins 1 to $N/2$	$N/2$	128
One axis	67	128
Three axes	201	128

- ▶ The block of the studio also gives the skewness and the kurtosis of the 16 power values. Add two values per axis when you need them.
- ▶ With no filter, the number of power values is half the FFT length.

Source: Values of the example of this module. The rule "half the FFT length" is from "Machine Learning Systems", chapter "DSP Spectral Features".

The choice of the FFT length

FFT length	Bins, distance	Values, 3 axes	Parameters
32	16, 1.56 Hz	57	1414
64	32, 0.78 Hz	105	2374
128	64, 0.39 Hz	201	4294

- ▶ The columns are one network with 20, 10, and 4 units, trained on the fallback dataset of the lab.
- ▶ Choose the length from the lowest frequency that separates two classes. Then check that the extra values still pay for themselves.

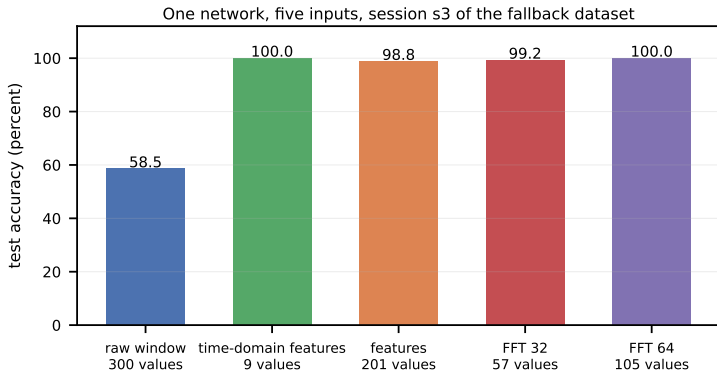
Source: Example model of this module, measured by the notebook of the lab on an x86 processor with AVX2. The values say nothing about a board.

- ▶ The FFT of a window gives one spectrum for the whole window. A shock in the first half and a shock in the second half give the same spectrum.
- ▶ The wavelet transform cuts the signal into a low part and a high part. Repeating the cut gives more levels.
- ▶ Each level is a signal of its own, and the same features apply to it.

With one level and no filter, 14 values for the signal and 14 for the low part, so 28 per axis and 84 for the three axes.

Feature group	Values
Statistics: n5, n25, n75, n95, median, mean, std, var, rms, skewness, kurtosis	11
Zero crossing rate and mean crossing rate	2
Entropy	1
One level	14

Source: "Machine Learning Systems", chapter "DSP Spectral Features", section "Wavelets".



The network on the raw window fails on this dataset: 1312 training windows are too few for 300 inputs. The nine time-domain values are enough. The 192 power values add nothing here.

Source: One run of the lab notebook on the fallback dataset of the module, an x86 processor with AVX2. The signals are simulated.

1. A window of 2 s at 50 Hz gives 100 samples per axis. Cut the recording with a stride, so the windows overlap.
2. Remove the mean first. Then compute the RMS, the skewness, and the kurtosis, and write the formula.
3. The FFT length sets the number of bins and the distance between two bins. Padding adds no information.
4. Keep the largest power of each bin over the frames when the motion has short events.
5. The feature set is a trade-off. Measure the accuracy and the model size before you choose it.

A window of 2 s is recorded at 50 Hz. You need the power at every frequency down to 0.4 Hz, and you want the fewest values that still reach that frequency.

1. Give the FFT length.
2. Give the number of values of one axis, with the three time-domain features.
3. Give the number of values of the vector of the three axes.

1. The distance between two bins is f_s/N , so $50/N \leq 0.4$ gives $N \geq 125$. The next power of two is $N = 128$. A window of 2 s cannot resolve less than 0.5 Hz, so 128 bins are enough for the grid and the data.
2. Bin 0 carries the mean, which is zero after the removal, so the answer starts at bin 1. The power values are $128/2 = 64$, and with the three time-domain features one axis has $3 + 64 = 67$ values.
3. The three axes have $3 \times 67 = 201$ values. A network with 20, 10, and 4 units then has 4294 parameters.

The answer of part 1 is the smallest FFT length that reaches 0.4 Hz. Whether 201 values are worth their place is a question of measurement, not of arithmetic: the lab measures it.

- ▶ **Goal.** You compute the features of a motion signal, train a small classifier on them, and compare five inputs for the same task.
- ▶ **Deliverable.** The table of section 8 of the notebook, copied into `report.md`, and the answers of its three questions.

Part	Content	Time
A	The data, the windows, and the three time-domain features	12 min
B	The FFT and the spectral power of one axis	13 min
C	The classifier, and its number of parameters	8 min
D	Five inputs for the same task, and the FFT length	12 min

File	Content
<code>spectral_features.ipynb</code>	The notebook. It has four tasks.
<code>host/motion_sim.py</code>	The simulated signals.
<code>host/make_dataset.py</code>	Makes the fallback dataset.
<code>report.md</code>	The report to hand in.
<code>solutions/</code>	The notebook with all tasks complete and with its output, and an example report.

- ▶ The notebook reads the folder `data/` when it has recordings of the format of the course. Otherwise it makes the simulated dataset in `fallback_data/`.
- ▶ The last session is the test session. A model that separates the sessions has learned the person, not the motion.

1. Start Jupyter from the folder of the lab:

```
../../../../.venv/bin/jupyter lab spectral_features.ipynb
```

2. Run section 1. Write the number of training windows and of test windows in the report.
3. **Task 1.** Write `time_features(window)`: the RMS, the skewness, and the kurtosis of one axis, after the removal of the mean.
4. Run the cell after the task.

You see the check of the task, and the three values for the first window:

RMS	Skewness	Kurtosis
0.0038	-0.3266	-0.0771

The first window is an idle window, so its values are small. That is the answer of the check, not a target for your own window.

Source: Values of one run of the solution notebook on the fallback dataset, an x86 processor with AVX2.

1. Run section 3. The cell draws the spectral power of one window of each class on the `z` axis. The `z` axis of `terrestrial` is almost still, because the motion of a truck sits in the horizontal plane. Write in the report which frequency holds the power of the class `lift`, and which one holds the power of `maritime`.
2. **Task 2.** Write `axis_features(window, nfft)`: the three time-domain features, then the power of bins 1 to `nfft // 2`.
3. Run the cell after the task, then section 4.

You see the number of values of one axis, and it must be $3 + 128/2 = 67$.

Then the feature matrix has 201 columns and the notebook prints the mean and the standard deviation of the first nine features.

The first three values of the feature vector are the RMS, the skewness, and the kurtosis of the axis '`x`'. The next six are power values.

1. **Task 3.** Write `make_model(n_inputs)`: a layer of 20 units with ReLU, a layer of 10 units with ReLU, and 4 outputs with softmax.
2. Run the cell after the task. The check calculates the expected number of parameters itself.
3. Run the next cell to train it.

input values 201, parameters 4294 (expected 4294), outputs 4
the model on the spectral features: 4294 parameters,
test accuracy 98.8 percent

The names of the layers and of the input matter. Without them, the file of the model changes size with each model that the notebook builds.

Source: Values of one run of the solution notebook on the fallback dataset, an x86 processor with AVX2.

Part D: five inputs for the same task (12 min)

1. Run section 6. It trains the same network on the raw window, on the time-domain features, and on the spectral features.
2. Write the three lines of output in the report.
3. **Task 4.** Before you run section 7, complete the table with the number of values of one axis for the FFT lengths 32, 64, and 128. Then write the length that you expect to win, and why.
4. Run section 7 and section 8, and copy the table into the report.

The notebook prints the same table as the figure of the lecture: the raw window reaches 58.5 percent, the nine time-domain values reach 100.0 percent, and the spectral values reach 98.8 percent with an FFT length of 128.

Source: Values of one run of the solution notebook on the fallback dataset, an x86 processor with AVX2. The signals are simulated.

- ▶ The notebook prints complete for tasks 1, 2, 3, and 4.
- ▶ The table of the report has five rows, each with the number of values, the number of parameters, and a measured test accuracy.
- ▶ The three questions of the report have answers with numbers from your own run.
- ▶ The Decision Log names one decision and one number.

Key point: Question of the Decision Log: a device must classify the motion of a pallet for a year on a battery. Which feature set do you put in the firmware, and which number of your table decides it?

Problem	Fix
The notebook says that it uses the fallback dataset	Your group has no recording in data/. The signals are simulated.
Task 2 says that the answer needs 67 values	Your answer has a different number. Return one value for each bin.
Task 3 says that the model raises an error	A layer is missing, or a layer is added with a tensor instead of with its input.
The accuracies change between two runs	Write the seed, and run the same cells again.

The README of the lab folder has the full list of the checks of the notebook and the run time.

- ▶ *Machine Learning Systems* by Vijay Janapa Reddi and contributors, Harvard University. mlsysbook.ai, CC BY-NC-SA 4.0.